

Chapter 4: Simple Linear Regression

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Deterministic Models

- Many problems in engineering and the sciences involve a study or analysis of the relationship between two or more variables.
- For example, the velocity of water in an open channel is related to the width of the channel, and the displacement of a particle at a certain time is related to its velocity.
 - If we let d_0 be the displacement of the particle from the origin at time $x = 0$ and v be the velocity,
 - the displacement at time x is $d_x = d_0 + vx$.

This is an example of a **deterministic** linear relationship because (apart from measurement errors) the model predicts displacement perfectly.

Deterministic Models

- However, in many situations, the relationship between variables is not deterministic.
- For example, the electrical energy consumption of a house (y) is related to the size of the house (x , in square feet), but it is unlikely to be a **deterministic** relationship.
- Similarly, the fuel usage of an automobile (y) is related to the vehicle weight x , but the relationship is **not deterministic**.
- In both of these examples, the value of y cannot be predicted perfectly from knowledge of the corresponding x .
- It is possible for different automobiles to have different fuel usage even if they weigh the same, and it is possible for different houses to use different amounts of electricity even if they are the same size.

Other examples

Other examples of variables related in a nondeterministic fashion include

- x = age of a child and y = size of that child's vocabulary,
- x = size of an engine (cm^3) and y = fuel efficiency for an automobile equipped with that engine,
- x = applied tensile force and y = amount of elongation in a metal strip.

Empirical Models

- The collection of statistical tools that are used to model and explore relationships between variables that are related in a **non-deterministic** manner is called **regression analysis**. Because problems of this type occur so frequently in many branches of engineering and science, regression analysis is one of the most widely used statistical tools.
- For example, in a chemical process, suppose that the yield of the product is related to the process-operating temperature.
 - Regression analysis can be used to build a model to predict yield at a given temperature level.
 - This model can also be used for process optimization, such as finding the level of temperature that maximizes yield, or for process control purposes

Regression analysis

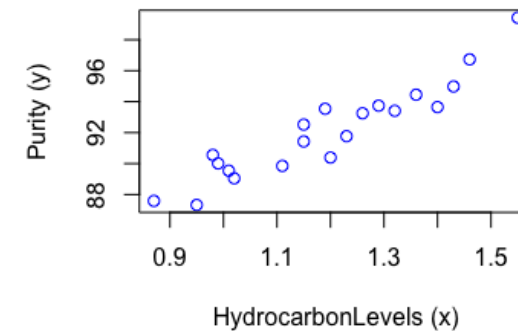
Regression analysis is the part of statistics that investigates the relationship between two or more variables related in a **non-deterministic** fashion. In this chapter, we

- generalize the **deterministic** linear relation $y = \beta_0 + \beta_1 x$ to a linear probabilistic relationship,
- develop procedures for making various inferences based on the model,
- obtain a quantitative measure (the correlation coefficient) of the extent to which the two variables are related.

Example 1

	HydrocarbonLevels	Purity
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

Example 1



- The simplest **deterministic** mathematical relationship between two variables x and y is a linear relationship $y = \beta_0 + \beta_1 x$.
- The objective of this section is to develop a **linear probabilistic model**.
- If the two variables are not deterministically related, then for a fixed value of x , there is uncertainty in the value of the second variable.

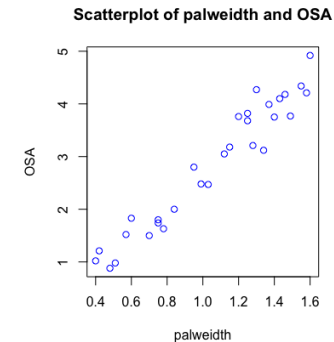
Example:

- If we are investigating the relationship between age of child and size of vocabulary and decide to select a child of age $x = 5.0$ years, then before the selection is made, vocabulary size is a **random variable Y**.
- After a particular 5-year-old child has been selected and tested, a vocabulary of 2000 words may result.
- We would then say that the observed value of Y associated with fixing $x = 5.0$ was $y = 2000$.

Example 2

Data: $n = 30$

- y : OSA (cm^2), ocular surface area
- x : width of the palprebal fissure (cm) (i.e., the horizontal width of the eye opening)



Linear Probabilistic Model

- The **deterministic model**

$$y = \beta_0 + \beta_1 x$$

the actual observed value of y is a linear function of x . The appropriate generalization of this to a **probabilistic model** assumes that the **expected value of Y is a linear function of x** , but that for fixed x the variable Y differs from its expected value by a random amount.

- The **Simple Linear regression Model**

$$Y = \beta_0 + \beta_1 x + \varepsilon \quad \text{where} \quad \varepsilon \sim \mathcal{N}(0; \sigma^2)$$

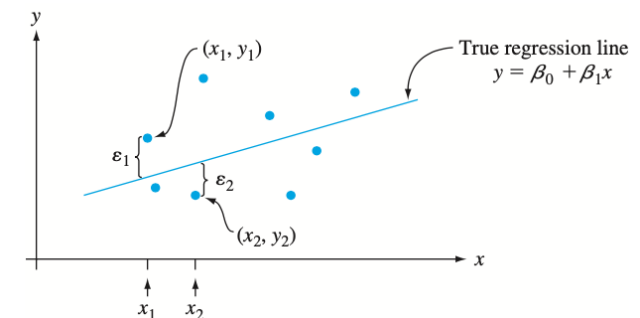
with

- β_0 and β_1 are parameters (intercept and slope).
- ε is a random error term which collects all the variation that cannot be explained by the linear relationship $y = \beta_0 + \beta_1 x$.

Simple Linear Regression

Given paired data $(x_1, y_1), \dots, (x_n, y_n)$, we have a linear stochastic relationship

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n.$$



Simple Linear Regression

At a fixed x , the distribution of Y : $Y \sim \mathcal{N}(\beta_0 + \beta_1 x, \sigma^2)$.

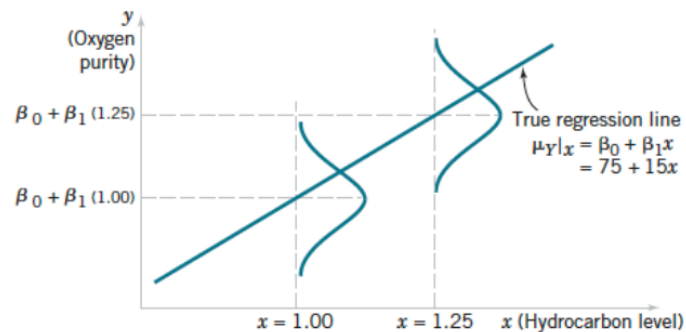
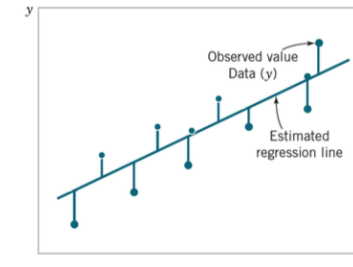


Fig: Distribution of Y for different values of x

- In simple linear regression, we have

$$Y = \beta_0 + \beta_1 x + \varepsilon \quad \text{where } \varepsilon \sim \mathcal{N}(0; \sigma^2)$$

- There are three unknown parameters β_0 (intercept), β_1 (slope), and σ^2 (the variance of the random error).
- We wish to estimate based on n pairs of observations (x_i, y_i) , $i = 1, \dots, n$. The estimation of β_0 and β_1 comes from the method called Least Squares.



Least Squares Estimates

- Vertical deviation of the point (x_i, y_i) from the line $y = \beta_0 + \beta_1 x$ is
height of point – height of line = $y - (\beta_0 + \beta_1 x)$

- Sum of squared vertical deviations from the points $(x_1, y_1), \dots, (x_n, y_n)$ to the line is then

$$f(\beta_0, \beta_1) = \sum_{i=1}^n [y - (\beta_0 + \beta_1 x)]^2.$$

- Estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ called **the least squares estimates**, are those values that minimize $f(\beta_0, \beta_1)$. That is, $\hat{\beta}_0$ and $\hat{\beta}_1$ are such that $f(\hat{\beta}_0, \hat{\beta}_1) \leq f(\beta_0, \beta_1)$ for any β_0 and β_1 .
- The **estimated regression line** or **least squares line** is then the line whose equation is $y = \hat{\beta}_0 + \hat{\beta}_1 x$.

Least Squares Estimates

- The least squares estimate of the slope coefficient β_1 of the true regression line is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}.$$

- The least squares estimate of the intercept β_1 of the true regression line is

$$\hat{\beta}_0 = \frac{\sum_{i=1}^n y_i - \hat{\beta}_1 \sum_{i=1}^n x_i}{n} = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Least Squares Estimates

The computational formulas for S_{xx} and S_{xy}

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - \frac{\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n}$$

Example 3: The cetane number of biodiesel fuels

- Data: $n = 14$ biofuels, x : iodine value (g) and y : cetane number

x	132.0	129.0	120.0	113.2	105.0	92.0	84.0	83.2	88.4	59.0	80.0	81.5	71.0	69.2
y	46.0	48.0	51.0	52.1	54.0	52.0	59.0	58.7	61.6	64.0	61.4	54.6	58.8	58.0

- The intercept and slope of the least squares line

```
> #Least Squares Estimates
> Sxx = sum((x - mean(x)) ^ 2)
> Sxy = sum((x - mean(x)) * (y - mean(y)))
> c(Sxx, Sxy)
[1] 6802.769 -1424.414
> beta_1_hat = Sxy / Sxx
> beta_0_hat = mean(y) - beta_1_hat * mean(x)
> c(beta_0_hat, beta_1_hat)
[1] 75.2124319 -0.2093874
> #use the lm() function to check the work we had previously done.
> mod = lm(y ~ x)
> coef(mod)
(Intercept)          x
75.2124319   -0.2093874
```

Scatterplot with least squares line superimposed, from R.

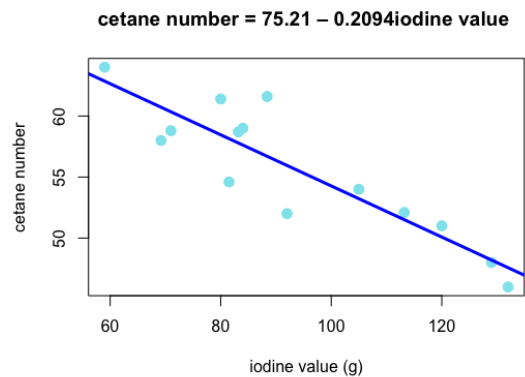


Fig1: Scatterplot of the data with the least squares line superimposed

Making Predictions

The estimated regression line can immediately be used for two different purposes. For a fixed x value x_0 , $y_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$ (the height of the line above x_0) gives either

- a point estimate of the expected value of Y when $x = x_0$
- a point prediction of the Y value that will result from a single new observation made at $x = x_0$

Example: Predict responses for new observations

- The estimated regression equation was $y = 75.212 - 0.2094x$. A point estimate of true average cetane number for all biofuels whose iodine value is $x_0 = 100$ is

$$\hat{\mu}_{Y|x=100} = \hat{\beta}_0 + \hat{\beta}_1 x_0 = 75.212 - 0.2094 * 100 = 54.27$$

If a single biofuel sample whose iodine value is 100 is to be selected, 54.27 is also a point prediction for the resulting cetane number.

- Using predict() function in R

```
> #Point prediction of the Y value that will result from a single new observation made at x=x_0=100
> predict(mod, newdata = data.frame(x = 100))
      1
54.27369
> #predict multiple values at once
> predict(mod, newdata = data.frame(x = c(90,100, 120)))
      1          2          3
56.36756 54.27369 50.08594
```

Residuals

- The fitted (or predicted) values $\hat{y}_1, \dots, \hat{y}_n$ are obtained by successively substituting $x_1, \dots, x_n$ into the equation of the estimated regression line:

$$\hat{y}_1 = \hat{\beta}_0 + \hat{\beta}_1 x_1, \quad \dots, \quad \hat{y}_n = \hat{\beta}_0 + \hat{\beta}_1 x_n$$

- The residuals are the differences

$$e_1 = y_1 - \hat{y}_1, \quad \dots, \quad e_n = \hat{y}_n - y_n$$

between the observed and fitted y values.

Example 4

- Data
 - $n = 20$
 - x : the machine's filtration rate (kg-DS/m/hr)
 - y : the moisture content of compressed pellets (%)
- Equation of least squares line: $y = 72.96 + 0.041x$.
- Fitted values are calculated from $\hat{y}_i = 72.96 + 0.041x_i$:

$$\hat{y}_1 = 72.96 + 0.041 * 125.3 \approx 78.100.$$

- Residual $e_1 = y_1 - \hat{y}_1 \approx 77.9 - 78.1 \approx -0.200$.

Output in R:

```
> # linear model
> mod = lm(moistcon ~ filtrate)
> mod
Call:
lm(formula = moistcon ~ filtrate)

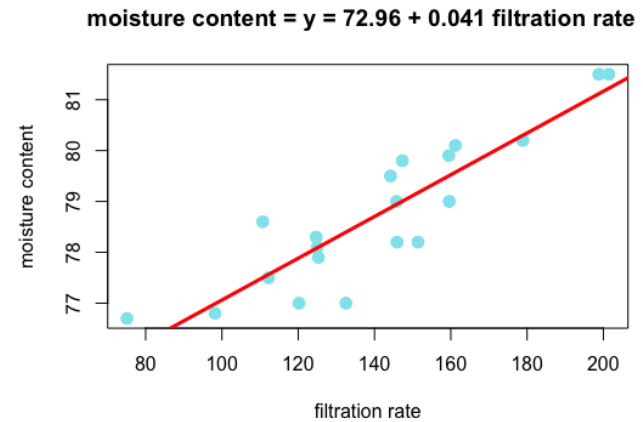
Coefficients:
(Intercept)    filtrate 
  72.95855      0.04103 

> ##return the coefficients
> coef(mod)
(Intercept)    filtrate 
72.95854697    0.0410377
```

Output in R:

```
> # return residuals, and fitted values
> Residual = mod$residuals # Residual = resid(mod)
> fit=fitted(mod) #fit = predict(mod)
> s2_e = sum(e^2) / (n - 2)
> data = xmp12.06
> fit=round(fit,digits=3)
> Residual=round(Residual,digits=3)
> restab=data.frame(data,fit, Residual)
> restab
```

	moistcon	filtrate	fit	Residual
1	77.9	125.3	78.100	-0.200
2	76.8	98.2	76.988	-0.188
3	81.5	201.4	81.223	0.277
4	79.8	147.3	79.003	0.797
5	78.2	145.9	78.945	-0.745
6	78.3	124.7	78.075	0.225
7	77.5	112.2	77.563	-0.063
8	77.0	120.2	77.891	-0.891
9	80.1	161.2	79.573	0.527
10	80.2	178.9	80.299	-0.099
11	79.9	159.5	79.503	0.397
12	79.0	145.8	78.941	0.059
13	76.7	75.1	76.040	0.660
14	78.2	151.4	79.171	-0.971
15	79.5	144.2	78.876	0.624
16	78.1	125.0	78.088	0.012
17	81.5	198.8	81.116	0.384
18	77.0	132.5	78.396	-1.396
19	79.0	159.6	79.508	-0.508
20	78.6	110.7	77.501	1.099



Nine of the 20 residuals are negative, so the corresponding nine points in a scatterplot of the data lie below the estimated regression line.

Estimate of σ^2

- The **error sum of squares** (equivalently, residual sum of squares), denoted by SSE, is

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n [y_i - (\hat{\beta}_0) + \hat{\beta}_1 x_i]^2 = S_{yy} - \hat{\beta}_1 S_{xy}.$$

- The estimate of σ^2 is

$$\hat{\sigma}^2 = s^2 = \frac{SSE}{n-2} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}.$$

Example

Back to Example 4, the residuals for the filtration rate and moisture content data:

- The corresponding error sum of squares is

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = (-0.2)^2 + \cdots + (1.099)^2 = 7.968.$$

- The estimate of σ^2 is

$$\hat{\sigma}^2 = s^2 = \frac{SSE}{n-2} = \frac{7.968}{20-2} = 0.4427$$

- and the estimated standard deviation is

$$\hat{\sigma} = s = \sqrt{0.4427} = 0.665.$$

Roughly speaking, 0.665 is the magnitude of a typical deviation from the estimated regression line—some points are closer to the line than this and others are farther away.

Example 5

How the propagation of an ultrasonic stress wave through a substance depends on the properties of the substance.

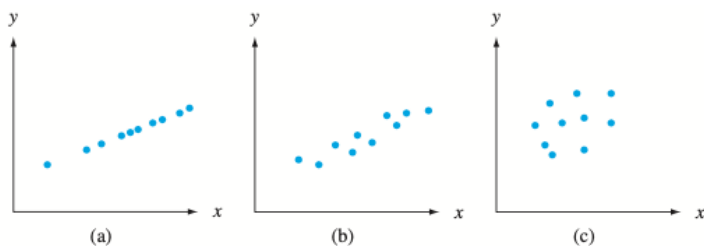
- Data
 - $n = 14$
 - x : fracture strength (as a percentage of ultimate tensile strength)
 - y : attenuation (in neper/cm, the decrease in amplitude of the stress wave) in fiberglass-reinforced polyester composites.
- Equation of least squares line: $y = 3.6209072 - 0.0147109x$.
- $SSE = S_{yy} - \hat{\beta}_1 S_{xy} = 103.54 - (37.6)^2/14 - (-0.0147109)(-155.98571429) = 0.2624532$.
- $\hat{\sigma}^2 = s^2 = \frac{SSE}{n-2} = \frac{0.2624532}{14-2} = 0.0218711$ and $s = 0.1479$.

Output in R:

```
> # linear model
> mod8 = lm(xmp12.08$y ~ xmp12.08$x)
> coef(mod8)
(Intercept) xmp12.08$x
3.62090564 -0.01471088
> # residuals
> e = resid(mod8)
> #estimate for sigma^2
> n=14
> s2_est = sum(e^2) / (n - 2)
> s2_est
[1] 0.02187137
> #residual standard error
> s_est = sqrt(s2_est)
> s_est
[1] 0.1478897
```

Error sum of squares (SSE)

The **SSE** can be interpreted as a measure of how much variation in y is left unexplained by the model - that is, how much cannot be attributed to a linear relationship.

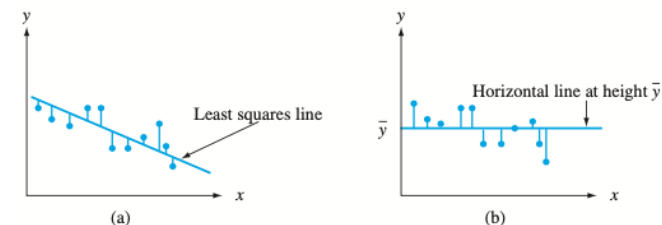


In Figure (a), $SSE = 0$, and there is no unexplained variation, whereas unexplained variation is small for the data of Figure (b) and much larger in Figure (c).

Total sum of squares (SST)

A quantitative measure of the total amount of variation in observed y values is given by the **total sum of squares (SST)**

$$SST = S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2 / n$$



Sums of squares illustrated:

- Fig (a): SSE = sum of squared deviations about the least squares line;
- Fig (b): SST = sum of squared deviations about the horizontal line.

Coefficient of determination

- The **coefficient of determination**, denoted by r^2 , is given by

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SST - SSE}{SST} = \frac{SSR}{SST} \quad \text{with} \quad 0 \leq r^2 \leq 1$$

where $SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = SST - SSE$.

- It is interpreted as the proportion of observed y variation that can be explained by the simple linear regression model (attributed to an approximate linear relationship between y and x).
- $r^2 \rightarrow 1$: the more successful is the simple linear regression model in explaining y variation.
- $r^2 \rightarrow 0$: an analyst will usually want to search for an alternative model (either a nonlinear model or a multiple regression model) that can more effectively explain y variation.

Example: The cetane number of biodiesel fuels

Back to Example 3,

- The scatterplot of the iodine value - cetane number data in Figure 1 (21) portends a reasonably high r^2 value.

- We have

$$SST = 43745.22 - (779.2)^2/14 = 377.174$$

$$SSE = 43745.22 - 75.212432 \cdot 779.2 - (-0.20938742 \cdot 71347.30) = 78.920$$

- The coefficient of determination is then

$$r^2 = 1 - \frac{SSE}{SST} = 1 - \frac{78.920}{377.174} = 0.791.$$

That is, 79.1% of the observed variation in cetane number is attributable to (can be explained by) the simple linear regression relationship between cetane number and iodine value.

Calculation in R

Back to Example 3 - The cetane number of biodiesel fuels with

- Data: $n = 14$ biofuels, x: iodine value (g) and y: cetane number
- The data can be fitted as following

```
> mod=lm(cetane_number~ iodine_value)
> mod

Call:
lm(formula = cetane_number ~ iodine_value)

Coefficients:
(Intercept)  iodine_value
    75.2124      -0.2094

> # mod$fitted.values      # compute the fitted values
> # mod$residuals          # compute all the residuals
> lm.est(mod)
      Estimate
beta0 75.2124319
beta1 -0.2093874
sigma  2.5644990
```

Calculation in R (continued)

```
> summary(mod)

Call:
lm(formula = cetane_number ~ iodine_value)

Residuals:
    Min       1Q   Median       3Q      Max
-3.9488 -1.5665  0.6817  1.0846  4.8974

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  75.21243    2.98363   25.208 9.22e-12 ***
iodine_value -0.20939    0.03109   -6.734 2.09e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.564 on 12 degrees of freedom
Multiple R-squared:  0.7908,    Adjusted R-squared:  0.7733
F-statistic: 45.35 on 1 and 12 DF, p-value: 2.091e-05
```

Example 6:

Data: $n = 14$ with x : burner-area liberation rate (MBtuyhr-ft²) and y : NO_x emission rate (ppm). The data can be read and fitted as following

```
> exm12_10=read.table("ex12.19.txt", header = TRUE)
> dim(exm12_10)
[1] 14 2
> mod10=lm(emission ~area)
> summary(mod10)

Call:
lm(formula = emission ~ area)

Residuals:
    Min       1Q   Median       3Q      Max
-77.88 -26.20  5.23  24.12  47.69

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -45.55191   25.46779  -1.789   0.0989 .
area          1.71143    0.09969   17.168 8.23e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 36.75 on 12 degrees of freedom
Multiple R-squared:  0.9609,    Adjusted R-squared:  0.9576
F-statistic: 294.7 on 1 and 12 DF,  p-value: 8.226e-10
```

Example 6

- Suppose that the slope and intercept of the true regression line are $\beta_1 = 1.70$, $\beta_0 = -50$ with $\sigma = 35$ (consistent with the values $\hat{\beta}_1 = 1.7114$, $\hat{\beta}_0 = -45.55$, $s = 36.75$).
- We proceeded to generate a sample of random deviations $\tilde{\varepsilon}_1, \dots, \tilde{\varepsilon}_n$ from a normal distribution $\mathcal{N}(0; 35^2)$.
- Then, we added $\tilde{\varepsilon}_i$ to $\beta_0 + \beta_1 x_i$ to obtain 14 corresponding y values.
- Regression calculations were then carried out to obtain the estimated slope, intercept, and standard deviation.
- This process was repeated a total of 20 times, resulting in the values given in following table.

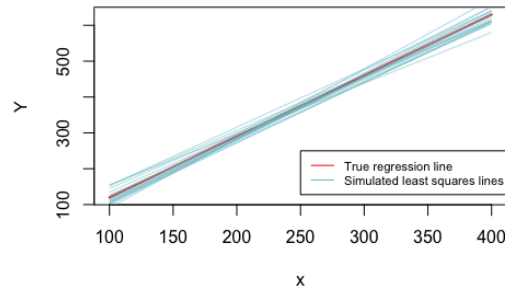
Calculation in R

```
> #Simulation Results for Example
> tab = data.frame()
> for(i in 1:20){
+   eps = rnorm(14,mean=0,sd=35)
+   y = -50+1.7*area+eps
+   mod=lm(y~area)
+   result=summary(mod)
+   tab = rbind(tab,c(mod$coefficients,result$sigma))
+ }
> names(tab) = c("beta1.hat","beta0.hat","s")
>
> #Graphs of the true regression line and 20 least squares lines
> plot(x=area,y=-50+1.7*area,type='l',col='red',lwd=2, xlab="x", ylab="Y")
> for(i in 1:20) lines(x=area,y=tab[i,2]*area+tab[i,1],col='cadetblue3',lty=1,lwd=0.5)
> legend(250, 250, legend=c("True regression line", "Simulated least squares lines"),
+       col=c("red", "cadetblue3"), lty=c(1,1), cex=0.7)
```

```
> tab
      beta1.hat beta0.hat      s
1      12.962303  1.417932 33.43938
2     -73.946431  1.767952 30.91510
3     -37.855969  1.625983 35.38584
4     -58.793668  1.743181 54.83400
5     -57.258321  1.772791 25.44049
6     -65.530244  1.687782 31.27833
7     -10.316343  1.553679 35.16707
8     -63.303532  1.725059 26.05918
9     -62.225434  1.677598 15.47662
10    -62.154087  1.730298 35.15555
11    -33.558660  1.597462 21.67200
12   -103.254384  1.939780 35.08561
13    -43.607221  1.660297 38.65372
14    -68.061003  1.773817 39.00273
15     -9.036423  1.624637 22.01641
16    -24.920224  1.595648 29.95608
17    -70.010275  1.757996 47.68424
18    -1.334530  1.540872 24.14076
19   -48.947055  1.633639 33.53088
20   -86.930468  1.845771 51.27851
```

Remarks

- There is clearly variation in values of the estimated slope and estimated intercept, as well as the estimated standard deviation.
- The equation of the least squares line thus varies from one sample to the next.
- The graphs of the true regression line and the 20 sample regression lines.



- The mean value of $\hat{\beta}_1$ is

$$E[\hat{\beta}_1] = \mu_{\hat{\beta}_1} = \beta_1$$

so $\hat{\beta}_1$ is an unbiased estimator of β_1 .

- The variance and standard deviation of $\hat{\beta}_1$ are

$$V[\hat{\beta}_1] = \sigma_{\hat{\beta}_1}^2 = \frac{\sigma^2}{S_{xx}} \quad \text{ans} \quad \sigma_{\hat{\beta}_1} = \frac{\sigma}{\sqrt{S_{xx}}}$$

The estimated standard deviation, i.e., estimated standard error, of $\hat{\beta}_1$:

$$\hat{\sigma}_{\hat{\beta}_1} = s_{\hat{\beta}_1} = \frac{s}{\sqrt{S_{xx}}}$$

- The estimator $\hat{\beta}_1$ has a normal distribution, i.e., $\hat{\beta}_1 \sim \mathcal{N}(\beta_1; \frac{\sigma^2}{S_{xx}})$.

- The assumptions of the simple linear regression model imply that the standardized variable

$$T = \frac{\hat{\beta}_1 - \beta}{s / \sqrt{S_{xx}}} = \frac{\hat{\beta}_1 - \beta}{s_{\hat{\beta}_1}}$$

has a t distribution with $n - 2$ df.

- A confidence Interval for β_1

- We begin with a probability statement:

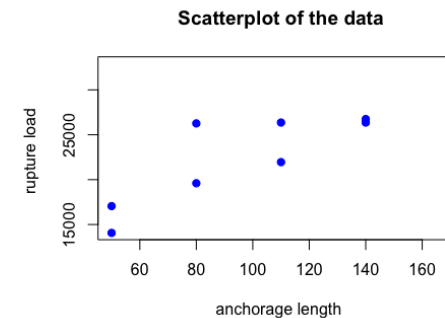
$$\mathbb{P}(-t_{\alpha/2, n-2} < \frac{\hat{\beta}_1 - \beta}{s_{\hat{\beta}_1}} < t_{\alpha/2, n-2}) = 1 - \alpha.$$

- A $100(1 - \alpha)\%$ CI for the slope β_1 of the true regression line is

$$\hat{\beta}_1 \pm t_{\alpha/2, n-2} * s_{\hat{\beta}_1}.$$

Example 7

Data: $n = 10$ with x : anchorage length (mm) and y : rupture load (N).



- Note that the relationship between anchorage length and rupture load is clearly not deterministic.
- There appears to be a rather substantial positive linear relationship between the two variables.

Calculation in R

```
> summary(mod)
```

```
Call:
lm(formula = y ~ x)
```

Residuals:

Min	1Q	Median	3Q	Max
-2817.5	-1548.4	-526.7	950.6	5674.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10698.33	2338.68	4.575	0.00181 **
x	123.64	19.84	6.233	0.00025 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2661 on 8 degrees of freedom
Multiple R-squared: 0.8293, Adjusted R-squared: 0.8079
F-statistic: 38.85 on 1 and 8 DF, p-value: 0.0002502

- A 95% confidence interval (CI) of β_1 is

$$\hat{\beta}_1 \pm t_{0.025/2,8} * s_{\hat{\beta}_1} = 123.64 \pm 2.306 * 19.84 = (77.90, 169.38)$$

- Calculation in R

```
> # CI
> confint(mod)

                2.5 %      97.5 %
(Intercept) 5305.33810 16091.3286
x           77.90054   169.3861
```

- Remarks

- With a high degree of confidence, we estimate that an increase in true average rupture strength of between 77.90 N and 169.38 N is associated with an increase of 1 mm in anchorage length (at least for anchorage lengths between 50 mm and 170 mm).
- The interval includes only positive values, so we can be quite confident of the tendency for strength to increase as anchorage length increases.

Hypothesis-testing Procedures

- Hypotheses:

$$\begin{cases} H_0 : \beta_1 = 0 \\ H_1 : \beta_1 \neq 0 \end{cases}$$

- When H_0 is true, the test statistic $T = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}} \sim T_{n-2}$ (t distribution with $n - 2$ df) and the test statistic value: $t = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}}$.
- At a significance level α , the null hypothesis H_0 will be rejected in favor of the alternative H_1 if the $t = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}}$ falls too far into either tail of the $t_{\alpha/2, n-2}$ curve (resulting in a small p_{value}).
- Remark: p_{value} is the smallest value of α for which H_0 can be rejected. H_0 will be rejected if $p_{value} \leq \alpha$.

Example 8

- Data: $n = 18$ with x : test track speed (km/h) and y : rolling test speed.
- Output in R

```
> summary(mod12)

Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-0.85464 -0.38796  0.09119  0.43084  0.71199

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.22414    3.52842  -0.63   0.537
x             1.08342    0.07806  13.88 2.43e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5069 on 16 degrees of freedom
Multiple R-squared:  0.9233,    Adjusted R-squared:  0.9185
F-statistic: 192.7 on 1 and 16 DF,  p-value: 2.432e-10
```

Example 8 (continued)

- The R output gives $r^2 = 0.9233$ which certainly portends a useful linear relationship.
- Let's carry out the model utility test at a significance level $\alpha = 0.01$.
 - Hypotheses:

$$\begin{cases} H_0 : \beta_1 = 0 \\ H_1 : \beta_1 \neq 0 \end{cases}$$
 - From this output, we have $\hat{\beta}_1 = 1.08342$, $s_{\hat{\beta}_1} = 0.07806$ and the test statistic value $t = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}} = 13.88$ and $p_{value} = 2.43e^{-10}$.
 - $p_{value} = 2.43e^{-10} \ll \alpha$, therefore H_0 can be rejected at any reasonable significance level.

Regression and ANOVA

ANOVA Table for Simple Linear Regression

Source of Variation	Sum of Squares	df	Mean Square	f
Regression	SSR	1	$MSR = SSR/1$	$\frac{MSR}{MSE}$
Error	SSE	$n - 2$	$MSE = s^2 = \frac{SSE}{n-2}$	
Total	SST	$n - 1$		

F test procedures

- Hypotheses:

$$\begin{cases} H_0 : \beta_1 = 0 \\ H_1 : \beta_1 \neq 0 \end{cases}$$
- When H_0 is true, the test statistic $F = \frac{SSR}{SSE/(n-2)} \sim F_{1,n-2}$ (F distribution with 1 and $n - 2$ df) and the test statistic value: $f = \frac{SSR}{SSE/(n-2)}$.
- At a significance level α , the null hypothesis H_0 will be rejected in favor of the alternative H_1 if the $f > F_{\alpha;1,n-2}$ or H_0 will be rejected if $p_{value} \leq \alpha$.

Example 8 (continued)

- Output in R

```
> anova(mod12)
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value    Pr(>F)    
x             1  49.500   49.500   192.65 2.432e-10 ***
Residuals    16   4.111    0.257                
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- The ANOVA table at the top of the output has $f = 192.65$ with a $p_{value} = 2.432e^{-10}$ for the model utility test. Therefore H_0 can be rejected at any reasonable significance level.
- Remark: The F test gives exactly the same result as the model utility t test because $t^2 = f$ and $t_{\alpha/2,n-2}^2 = F_{\alpha;1,n-2}$. We have $t^2 = 13.88^2 = 192.65 = f$ and the same $p_{value} = 2.432e^{-10}$.

Example 7 (continued)

- Output in R

```
> anova(mod)
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value    Pr(>F)    
x             1  275178130  275178130   38.852 0.0002502 ***
Residuals    8  56661439    7082680
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- The ANOVA table at the top of the output has $f = 38.852$ with a $p_{value} = 0.0002502$ for the model utility test. Therefore H_0 can be rejected at any reasonable significance level.
- Remark: We have $t^2 = 6.233^2 = 38.85029 \approx 38.852 = f$ and the same $p_{value} = 0.00025$.

Let $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x_0$, where x_0 is some fixed value of x . Then

- The mean value of $\hat{Y}$ is

$$\mathbb{E}[\hat{Y}] = \mathbb{E}[\hat{\beta}_0 + \hat{\beta}_1 x_0] = \mu_{\hat{\beta}_0 + \hat{\beta}_1 x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

- The variance of $\hat{Y}$ is

$$\mathbb{V}[\hat{Y}] = \sigma_{\hat{Y}}^2 = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]$$

and the estimated standard deviation of $\hat{\beta}_0 + \hat{\beta}_1 x_0$ is

$$s_{\hat{Y}} = s_{\hat{\beta}_0 + \hat{\beta}_1 x_0} = s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

- $\hat{Y}$ has a normal distribution.

Inferences concerning $\mu_{Y|x_0}$ and PI for a future value of Y

- The variable

$$T = \frac{\hat{\beta}_0 + \hat{\beta}_1 x_0 - (\beta_0 + \beta_1 x_0)}{s_{\hat{\beta}_0 + \hat{\beta}_1 x_0}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{s_{\hat{Y}}}$$

has a t distribution with $n - 2$ df.

- A $100(1 - \alpha)\%$ CI for $\mu_{Y|x_0}$, the expected value of Y when $x = x_0$, is

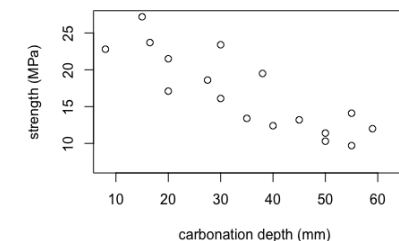
$$\hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{\alpha/2, n-2} s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} = \hat{y} \pm t_{\alpha/2, n-2} s_{\hat{Y}}$$

- A $100(1 - \alpha)\%$ Prediction Interval (PI) for a future Y observation to be made when $x = x_0$ is

$$\hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{\alpha/2, n-2} s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} = \hat{y} \pm t_{\alpha/2, n-2} \sqrt{s^2 + s_{\hat{Y}}^2}$$

Example 9: Data: $n = 18$, x : carbonation depth (mm) and y : strength (MPa). The data can be read and fitted as following.

```
> # Load dataset from Devore7 library
> library(Devore7)
> data("xmp12.13")
> #attach(xmp12.13)
> dim(xmp12.13)
[1] 18 2
> head(xmp12.13, 2)
   x    y
1  8 22.8
2 15 27.2
> plot(x.v, xlab="carbonation depth (mm)", ylab="strength (MPa)")
```



Calculation in R:

```
> mod13=lm(y~x)
> summary(mod13)

Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-4.1317 -2.0043 -0.7488  2.1366  5.1439

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 27.18294    1.65135   16.461 1.88e-11 ***
x           -0.29756    0.04116   -7.229 2.01e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.864 on 16 degrees of freedom
Multiple R-squared:  0.7656,    Adjusted R-squared:  0.7509
F-statistic: 52.25 on 1 and 16 DF,  p-value: 2.013e-06
```

The estimated variance is $s^2 = \hat{\sigma}^2 = 2.864^2 = 8.202496$. The estimated regression coefficients are $\hat{\beta}_0 = 27.18294$ and $\hat{\beta}_1 = -0.29756$. The fitted model is $\hat{y} = 27.18294 - 0.29756x$. We obtain $t = -7.229$ with a $p_{value} = 2.01e^{-6}$ for the model utility test $\Rightarrow H_0: \beta_1 = 0$ can be rejected at any reasonable significance level.

Calculation in R (continued):

```
> anova(mod13)
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value    Pr(>F)
x             1  428.62   428.62  52.253 2.013e-06 ***
Residuals    16  131.24    8.20
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The ANOVA table of the output has

- $SSE = 131.24$, $MSE = s^2 = 8.20$,
- $f = 52.253$ with a $p_{value} = 2.013e^{-6}$ for the model utility test. Therefore $H_0: \beta_1 = 0$ can be rejected at any reasonable significance level.

Using a 95% confidence level, with $\hat{y} = 27.18 - 0.2976 \cdot 45 = 13.79$, $s_{\hat{y}} = 2.8640 \sqrt{\frac{1}{18} + \frac{(45-36.611)^2}{4840.7778}} = 0.7582$, $s = 2.8640$ and $t_{0.025,16} = 2.120$ calculate: A 95% CI for $\mu_Y|_{45}$:

$$13.79 \pm 2.120 \cdot 0.7582 = 13.79 \pm 1.61 = (12.18, 15.40)$$

A 95% PI for a future Y observation when $x = 45$ mm

$$13.79 \pm 2.120 \sqrt{2.8640^2 + 0.7582^2} = 13.79 \pm 6.28 = (7.51, 20.07)$$

Output in R

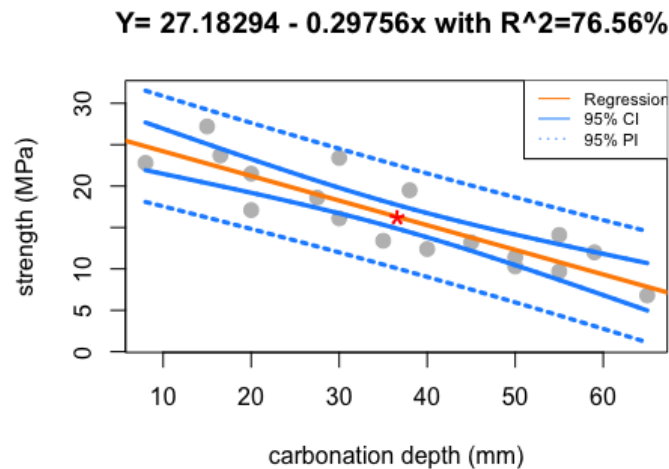
```
> #####confidence intervals for beta_0 and beta_1
> confint(mod13)
              2.5 %      97.5 %
(Intercept) 23.6822344 30.6836377
x           -0.3848254 -0.2102971
> #####Confidence interval for the mean response
> new_x = data.frame(x= 45)
> predict(mod13, newdata = new_x,interval = c("confidence"))
      fit      lwr      upr
1 13.79268 12.18525 15.40011
> #####Prediction Interval for New Observations
> predict(mod13, newdata = new_x, interval = c("prediction"))
      fit      lwr      upr
1 13.79268  7.512036 20.07333
```

Confidence and Prediction Bands

Calculation in R (continued):

```
#####Confidence and Prediction Bands
depth = seq(min(x), max(x), by = 0.01)
strength_CI_band = predict(mod13,
                           newdata = data.frame(x = depth),
                           interval = "confidence", level = 0.95)
strength_PI_band = predict(mod13,
                           newdata = data.frame(x = depth),
                           interval = "prediction", level = 0.95)
plot(y ~ x, data = xmp12.13,
     xlab = "carbonation depth (mm)",
     ylab = "strength (MPa)",
     main = "Y= 27.18294 - 0.29756x with R^2=76.56% ",
     pch = 20,cex = 2,col = "grey",
     ylim = c(min(strength_PI_band), max(strength_PI_band)))
abline(mod13, lwd = 3, col = "darkorange")
lines(depth, strength_CI_band[, "lwr"], col = "dodgerblue", lwd = 3, lty = 1)
lines(depth, strength_CI_band[, "upr"], col = "dodgerblue", lwd = 3, lty = 1)
lines(depth, strength_PI_band[, "lwr"], col = "dodgerblue", lwd = 3, lty = 3)
lines(depth, strength_PI_band[, "upr"], col = "dodgerblue", lwd = 3, lty = 3)
points(mean(x), mean(y), pch = "**", cex = 2, col="red")
legend(51, 33, legend=c("Regression", "95% CI", "95% PI"),
      col=c("darkorange", "dodgerblue", "dodgerblue"), lty=c(1,1,3), cex=0.7)
```

Figure scatterplot with CI and PI for the data of Example



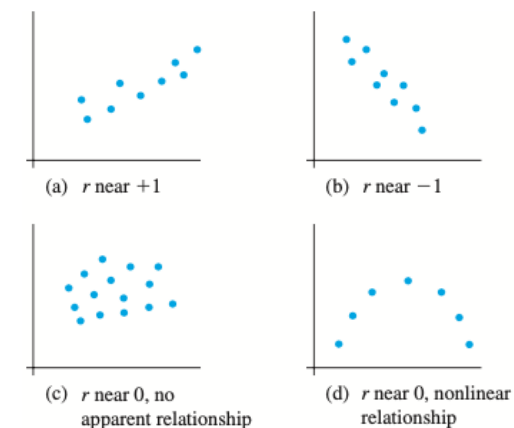
- The sample correlation coefficient for the n pairs $(x_1, y_1), \dots, (x_n, y_n)$ is

$$r = \frac{S_{xy}}{\sqrt{S_{xx}} \sqrt{S_{yy}}}$$

- The most important properties of r are as follows:
 - The value of r does not depend on which of the two variables under study is labeled x and which is labeled y .
 - The value of r is independent of the units in which x and y are measured.
 - $-1 \leq r \leq 1$.
 - $r = 1$ iff all (x_i, y_i) pairs lie on a straight line with positive slope, and $r = -1$ iff all (x_i, y_i) pairs lie on a straight line with negative slope.
 - The square of the sample correlation coefficient gives the value of the coefficient of determination, $(r)^2 = r^2$.

Remarks

- $r = 1$: the largest possible degree of positive relationship;
- $r = -1$: the most negative relationship;
- r measures the degree of linear relationship among variables.
- $r \rightarrow 0$ is not evidence of the lack of a strong relationship, but only the absence of a linear relation.
- "When can it be said that there is a strong correlation between the variables, and when is the correlation weak?"
 - Weak: $-0.5 \leq r \leq 0.5$
 - Moderate: either $-0.8 < r < -0.5$ or $0.5 < r < 0.8$
 - Strong: either $r \geq 0.8$ or $r \leq -0.8$

Data plots for different values of r 

Inferences About the Population correlation coefficient

- The population correlation coefficient $\rho(X, Y)$ is

$$\rho = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{V}(X)\text{V}(Y)}}$$

- The sample correlation coefficient to make various inferences about ρ . In particular, r is a point estimate for ρ , and the corresponding estimator is

$$\hat{\rho} = R = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2} \sqrt{\sum(Y_i - \bar{Y})^2}}$$

Testing for the absence of correlation

Let R denote the sample correlation coefficient as a random variable (before data is obtained).

- When $H_0 : \rho = 0$ is true, the test statistic

$$T = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}} \quad \text{has a } t \text{ distribution with } n-2 \text{ df.}$$

- Alternative Hypothesis p_{value} determination

- $H_1 : \rho > 0$ Area under the t_{n-2} curve to the right of t
- $H_1 : \rho < 0$ Area under the t_{n-2} curve to the left of t
- $H_1 : \rho \neq 0$ $2 \times$ Area under the t_{n-2} curve to the right of $|t|$

- Data: $n = 10$, x : maternal age (yr) and y : baby's birth weight (g).
- Output in R

```
> r=cor(x,y)
> r
[1] 0.8836937
> cor.test(x,y)

Pearson's product-moment correlation

data:  x and y
t = 5.34, df = 8, p-value = 0.0006941
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.5727359 0.9723228
sample estimates:
cor
0.8836937
```

- The point estimate of ρ is $\hat{\rho} = r \approx 0.884$.
- A 95% CI for ρ : (0.5727, 0.9723)
- $H_0 : \rho = 0$ can be rejected at any reasonable significance level ($p_{\text{value}} = 0.0006941$).